Mazen Mohamed Abdelsalam

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Profile

Final-year Artificial Intelligence student at Cairo University focused on LLM applications and Retrieval-Augmented Generation (RAG). Built a LangChain and Groq (Llama-3.3) tool that generates pytest suites from Python source code, completed NVIDIA DLI training in generative AI and RAG, and is training in AI automation with n8n through DEPI. Solved 2000+ algorithmic problems, with top regional ICPC rankings and a Meta Hacker Cup Round 2 qualification. Seeking an AI engineering internship.

Education

Bachelor of Science in Artificial Intelligence *Sep 2023 – Expected Jun 2027*
Faculty of Computers and Artificial Intelligence, Cairo University

- **Department:** Artificial Intelligence
- **GPA: 3.42/4.0** Ranked **20th** out of **170+ students** in AI Department
- **Key Courses:** Machine Learning, Data Structures, Database Systems, Software Engineering, Computer Networks, Operating Systems, Computer Architecture, OOP, Web Technologies, Computational Cognitive Science

Obour STEM School *Sep 2020 – Jul 2023*
High School Graduate

Experience

Digital Egypt Pioneers Initiative (DEPI) Trainee *Jul 2026 – Present*
AI & Data Science – AI Automation Workflow with n8n Professional

- Completing 200+ hours of hands-on training focusing on end-to-end AI automation workflows, LLM application engineering, and intelligent process automation.
- Engineered automated agentic workflows integrating LangChain, vector retrieval (RAG), and external toolchains to streamline complex data-processing tasks.
- Collaborated in agile team sprints to build and deploy practical AI automation pipelines and capstone projects adhering to clean code and version control standards.

Projects

Automated Unit Testing Generator

- Built a Python CLI application using LangChain and the Groq API (Llama-3.3) to automatically generate reliable pytest scripts for any Python function.
- Used Python's `ast` library to parse source files locally, ensuring the LLM only processes valid, single-function code blocks to prevent errors and reduce API calls.
- Designed strict system prompts enforcing the Arrange, Act, Assert (AAA) testing standard and correct mocking of external dependencies.
- Created an interactive terminal menu using `pathlib` to automatically save generated test suites in the correct local directory.
- **Tools:** Python, LangChain, Groq API (Llama-3.3), Pytest, Prompt Engineering

HERE Cairo Geospatial Student Hackathon 2026 – Spatial Road Pattern Detection

- Designed and developed a solution to detect spatial patterns in large-scale geospatial datasets under hackathon time constraints.
- Applied feature extraction techniques to identify road networks using probe and Vehicle Probe Data (VPD).
- Processed and cleaned spatial data using GeoPandas and QGIS to enhance dataset quality and usability for downstream analysis.
- **Tools:** Python, Pandas, NumPy, GeoPandas, QGIS

Sorting System – Algorithm Benchmarking Tool

- Implemented **9** sorting algorithms with detailed iteration output and a template-based system supporting **4** data types (int, float, string, char).
- Integrated microsecond-precision runtime benchmarking to compare algorithm performance across varying dataset sizes.
- **Tools:** C++

Achievements

Competitive Programming – ICPC ID

- Solved **2000+ problems** in Algorithms and Data Structures; participated in **110+ online contests**
- Ranked **814th / 13,779** in Meta Hacker Cup 2025 – Qualified to Round 2, earned Meta T-Shirt *Nov 2025*
- Ranked **40th / 430+ teams** in 2025 ICPC Egyptian Collegiate Programming Contest *Aug 2025*
- Ranked **18th / 110+ teams** in 2024 ICPC Africa and Arab Collegiate Programming Championship *Dec 2024*
- Ranked **3rd / 300+ teams** in 2024 ECPC Qualifications – Day 5 *Jul 2024*
- **Silver Honour** – IYMC International Youth Math Challenge *Dec 2022*
- **Gold Award** – Design Thinking With Robotics and Computational Thinking International Competition (Grade 11/12) *Jun 2022*

Skills

Programming Languages: Python, C++, SQL, Java, JavaScript, HTML, CSS

Frameworks & Tools: LangChain, Pandas, NumPy, GeoPandas, Matplotlib, Scikit-learn, Pytest, Groq API, n8n, QGIS

Concepts: Machine Learning, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Automation, Agentic Workflows, Data Analysis, Feature Engineering, Algorithms, Data Structures, OOP, Software Engineering, Database Systems

Leadership & Volunteering

Data Structures and Algorithms Coach & Problem Setter *Sep 2024 – May 2025*
Faculty Community, Cairo University

- Mentored **100+ students** in data structures and algorithms, preparing them for ECPC and ACPC competitive programming contests.
- Collaborated with a team of problem-setters to author problems with full solutions and test case suites.

Certifications & Training

NVIDIA DLI: Generative AI *Feb 2026*
Information Technology Institute (ITI)
Topics: Generative AI, LLMs, RAG, Prompt Engineering, Applied AI Workflows

Huawei HCIA: Artificial Intelligence *Aug 2024 – Sep 2024*
Huawei ICT Academy, Cairo University
Topics: Supervised/Unsupervised Machine Learning, Deep Learning Architectures

Competitive Programming Training – Level 2 & 3 *Jan 2022 – Sep 2022*
Coach Academy
Topics: Graph Theory, Dynamic Programming, String Hashing, Number Theory, Segment Trees, Sparse Tables